

Fluid dynamics* for Family and Friends

*See reverse for some general astronomy/exoplanet definitions

“Fluids”/“hydrodynamic” - when particles are close enough that they can bounce into one another (“collisional”) and, as a result, move as a continuous unit like liquid does.

Examples: *water, air, magma, traffic, giant flocks of birds*

“Sound speed” - the finite speed a wave can travel through a medium. Set by how long a particle has to travel before it hits its neighbor, then that neighbor hits its neighbor and so on.

Examples: *The sound speed at Earth’s surface = 767 mph (e.g., you hear a thunder clap arriving long after lightning strikes) vs. in a swimming pool = 2,250 mph because the water particles are closer together than air particles.*

“Supersonic” - something moving faster than the sound speed. In this talk, gas moving supersonically outruns the gas behind it.

“Sonic point” - The height above a planet’s surface at which the escaping gas accelerates faster than the sound speed and the gas becomes supersonic.

An everyday example of fluids: The crowded gossip hallway

A student enters the crowded hallway at one end. They walk in a random direction and whisper the juicy gossip with the first person they run into. Both students with the gossip continue walking in random directions and whisper it to first person they run into. Each person who receives the gossip whispers it to the next person they run into and so on.

In this example, the students can be considered to be a **“fluid”** and moving **“hydrodynamically”**, because they can run into one another and transmit information. The **“sound speed”** is the rate at which the gossip reaches the end of the hallway. Let’s imagine that it takes 1 minute for the gossip to travel the 20 foot hallway. The sound speed in the student-fluid is then 20 feet/minute.

If the students that exit the doors at the far end of the hallway start walking briskly to class at a rate of 30 feet/minute. Then the door is the **“sonic point”** and those students are then **“supersonic.”** That means that the whispered gossip would never catch them! Likewise, any new gossip they learn cannot make it back to the crowded hallway because they are moving faster than everyone behind them.

Apply these concepts:

Using what you have learned, predict why “in space, no one can hear you scream.”

Hint: What would happen if our randomly wandering gossipy student entered a football field with only two other people in it?

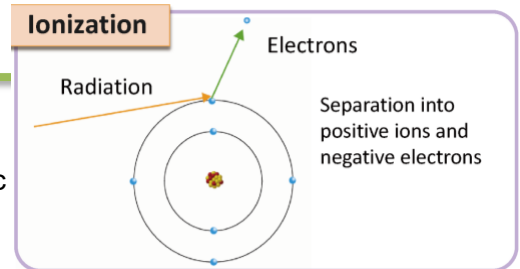
“Ionization” - Removal of an electron from an atom. For each element on the periodic table, each ring of electrons has a specific minimum energy required to remove an electron.

That energy is called the **“ionization energy”**.

In **“photoionization”** the electron is removed by a photon of UV or X-ray light.

In **“collisional/secondary ionization”** the electron is removed after a collision with a very fast moving electron freed during the initial photoionization described above.

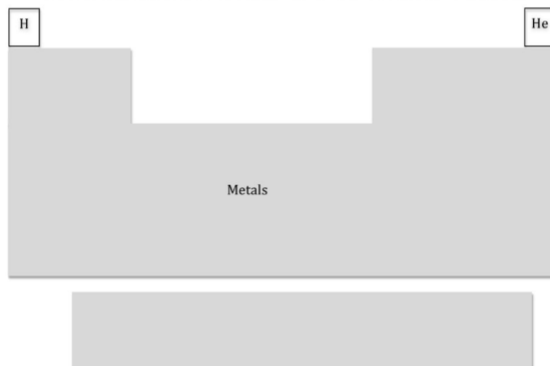
Examples: fluorescent lights and neon lights, Northern Lights



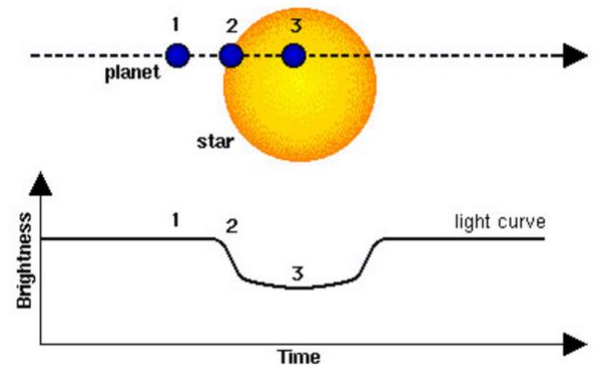
“Flux” - Amount of radiation per second that reaches the planet.

“Metal”

The Astronomer's Periodic Table



“Transit”



Wavelengths of light

Wavelength
(m)

Gamma ray
 10^{-12}

X-ray
 10^{-10}

Ultraviolet
 10^{-8}

Visible
 $.5 \times 10^{-6}$

Infrared
 10^{-5}

Microwave
 10^{-2}

Radio
 10^3



About the size of...



Atomic nuclei



Atoms



Molecules



Protozoans



Pinpoints



Honeybees



Humans



Buildings

Kelvin to Farenheit:

100 K → -280°F

1000 K → 1340°F (diamonds burn at 1600°F)

10,000 K → 17,500°F